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06/01/2025

ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: B. Tech. BT

Semester: III

Name of the Paper: Genetics & Molecular Biology

Paper Code: TBT 301

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub-questions among a, b & c in each main question
- (iii) The total marks for each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	State the role of various kinds of RNAs present in the cell.	CO1
(b)	Classify multiple gene theory. Compare Qualitative and Quantitative Genetics.	
(c)	Define the (i) X-linked inheritance (ii) Gene Linkage. How do they work in genetics?	
Q2	(20 marks)	
(a)	List the enzymes involved in DNA replication. State their functions.	CO2
(b)	Summarise different types of DNA damage and their causes.	
(c)	Discuss the structure, types and function of bacterial DNA polymerases.	
Q3	(20 marks)	
(a)	Illustrate how the mechanism of eukaryotic transcription is different from prokaryotic transcription.	CO3
(b)	Justify the importance of post-translational modification (PTM) in polypeptides and proteins. Citing examples, discuss how it is achieved.	
(c)	Differentiate between the prokaryotic & eukaryotic translation in 10 points.	
Q4	(20 marks)	
(a)	Evaluate the role of (i) constitutive and inducible gene expression (ii) positive and negative gene regulation.	CO4
(b)	Illustrate the mechanism of operation of Lactose operon in bacteria.	
(c)	Discuss the mechanism of gene regulation by histone modification in eukaryotic cells.	
Q5	(20 marks)	
(a)	Design protocol for polymerase chain reaction. Write a short note on its type & application.	CO5
(b)	Briefly discuss a few modern techniques to study gene expression.	
(c)	Define gene editing. How it is done efficiently by CRISPR cas-9.	

Note for the question paper setters:

- Question paper should cover all the COs of the course.

Joseph
Papineau